

ClariFi: Refinement Changelog

1. Problem Statement Update

After the preliminary round, key weaknesses in the delivering of problem statements were identified, tweaks and minor adjustments were made to the problem statement in order to make the audience understand personally the urgency of the problem.

The refined problem statement narrates around one central concept: Data darkness, pinpointing the fact that majority Malaysian consumers are actively enshrouded by this darkness and kept in blur from making sound spending decisions. This data darkness has led to unnotified money leakages and price anxiety, impacting their financial health.

2. Features Update

Feature	Preliminary Version	Refined Version
AI Fast Expense Capture	Voice or receipt photo capture, user reviews parsed result and confirms.	Unchanged. Core frictionless capture remains the foundation of the product.
Price Comparison Feature	- Converts captured bills and promotional brochures into price observations. - Single items lookup, where the user searches for one item, see prices at different stores	- Powered by direct partnerships with local groceries or supermarkets and crowdsourced receipts, building a living, real-time local price database. - Bucket Feature Implemented, users can now add multiple items into a list, and build a price matrix across all nearby stores to calculate the total basket cost at each location. It then weights results by proximity and recommends the single store where the user can buy everything at the most worthwhile combined price in one trip.
Expense Breakdown	View and filter recorded expenses.	Unchanged. Surfaces spending patterns and category-level leaks on a monthly basis.
Alerts & Notifications	Create threshold alerts and receive event notifications when conditions are met.	Unchanged. Proactively notifies users of price changes and budget thresholds.
AI Purchase Insights	<i>Not in preliminary version.</i>	AI analyses each purchase, flags overspending, directs users to cheaper alternatives via Community Price

		Intelligence, and explains price increases through real economy trends.
Family Profile	Share a family profile to track each other's spending and share promotions or deals.	Retained as a supporting feature. Enables shared household visibility and expense splitting.

3. Judges Feedbacks and Improvements

- **Feedback:** Stronger Technical Validation and Better AI Logic
Improvements: AI & Technology Stack

Layer	Technology Used	AI/ Logic	Output
Capture Intelligence	IOS speech recognition model (mobile), Gemini-3.1-flash-light (vision OCR), NestJS parsing pipeline	AI: Speech-to-text transcription + vision-based receipt reading. Logic: Named entity extraction pulls merchant, amount, date, category, and payment method from unstructured input. Low-confidence fields flagged for user review. Human-in-loop confirmation before saving ensures data integrity.	Structured, validated expense entry
Purchase Intelligence (AI Purchase Insights)	PostgreSQL price observation database, Gemini-3.1-flash-light (LLM reasoning), community receipt + grocery partnership data	AI: LLM generates a brief explanation of price deviations and economy-driven price trends. Logic: Price percentile comparison, where user's paid price is ranked against the distribution of the same item across nearby stores within a defined radius and recent time window. If price has risen broadly, time-series trend analysis identifies whether the increase is sustained or a one-time spike. Cheaper alternatives retrieved via geospatial query.	Spending verdict + cheaper alternatives + price trend explanation
Basket	PostgreSQL price	Logic: Price matrix built	Single store

Intelligence (Price Intelligence)	matrix query, `` v NestJS aggregation logic	across all key partners for every item in the list. Total basket cost calculated per store. Result weighted by proximity, where a further store must offer proportionally greater savings to rank higher. Stores ranked by best combined basket value and distance score.	recommendatio n with full basket breakdown and savings comparison
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